



THE UNIVERSITY
OF BRITISH COLUMBIA



Electrical and
Computer
Engineering

EECE 568: From Exploring to Building Machine Learning

ASSIGNMENT #1 – Math Review and Theoretical Questions

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EECE 568 - Assignment 1

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Due Date: October 1, 2026 - 11:59 pm

Late submissions will not be accepted. In case of any medical or family emergency, please inform the instructors.

Deliverables: Please upload a ZIP file to Canvas containing a PDF for section 1 and a Jupyter Notebook (.ipynb) for section 2.

1 Math Review and Theoretical Questions (50 marks)

1.1 Eigenvalue (6 marks)

For Matrices $A \in \mathbb{R}^{m \times n}$ and $B \in \mathbb{R}^{n \times m}$ Prove:

- If $m = n$, then A and A^T have the same set of eigenvalues
- AB and BA have the same set of eigenvalues

1.2 Rank (3 marks)

Prove that if $A \in \mathbb{R}^n$ is a full rank matrix, matrices B and $A^{-1}BA$ have the same set of eigenvalues.

1.3 Definite Matrix (3 marks)

A symmetric matrix $A \in \mathbb{R}^n$ is positive definite if $x^T Ax > 0$ for all non-zero vectors $x \in \mathbb{R}^n$. Prove if matrix A is positive definite, it is full rank and invertible.

1.4 Gram Matrix (6 marks)

Given any matrix $A \in \mathbb{R}^{m \times n}$, the matrix $G = A^T A$ is called a Gram matrix

Prove:

- G is always positive semidefinite
- If $m \geq n$ and A is full rank, then G is a positive definite matrix

1.5 Matrix Derivative (6 marks)

Consider the following function:

$$f(x, y) = \begin{bmatrix} f_1(x, y) \\ f_2(x, y) \\ f_3(x, y) \end{bmatrix} = \begin{bmatrix} x^2 y^3 \\ 4x^2 + \cos(y) \\ 4y^2 - 2x^2 \end{bmatrix}$$

- What are the dimensions of the Jacobian matrix of f using (row, column) format?

- What is $\frac{\partial f_2}{\partial y}$?
- What is $\frac{\partial^2 f_1}{\partial x \partial y}$?

1.6 Push-through Identity (3 marks)

Suppose A is $m \times n$, B is $n \times m$, and the $m \times m$ matrix $(\lambda I + AB)$ is invertible. Prove that the $n \times n$ matrix $(\lambda I + BA)$ is invertible. *Hint.* Show that $(\lambda I + BA)x = 0$ implies $(\lambda I + AB)y = 0$, where $y = Ax$.

1.7 Regression Coefficients (3 marks)

In building a linear regression model for a particular data set, you observe the coefficient of one of the features having a relatively high negative value. Which one is true and why?

- This feature has a strong effect on the model (should be retained)
- This feature does not have a strong effect on the model (should be ignored)
- It is not possible to comment on the importance of this feature without additional information

1.8 Independent Features (6 marks)

Given n training data with d features, let the target value vector be $Y = \begin{bmatrix} y^{(1)} \\ \dots \\ y^{(n)} \end{bmatrix} \in \mathbb{R}^n$ and data samples

be $X = \begin{bmatrix} x_1^{(1)}, \dots, x_d^{(1)} \\ \dots \\ x_1^{(n)}, \dots, x_d^{(n)} \end{bmatrix} \in \mathbb{R}^{n \times d}$. In this context, x_j denotes the j^{th} column of this matrix. Show that if

we train a regressor on just one of the features (from d features), we then have $w_j = \frac{x_j^T y}{x_j^T x_j}$, where w_j is the j^{th} element of the parameter vector.

1.9 Ridge Regression (5 marks)

Consider $X \in \mathbb{R}^{n \times d}$ as the training data with n samples and d features, and $y \in \mathbb{R}^n$ as the labels. Using the squared L2-norm with an L2-regularization, the minimization problem becomes:

$$\arg \min_{w \in \mathbb{R}^d} \frac{1}{2} \|Xw - y\|^2 + \frac{\lambda}{2} \|w\|^2$$

Where λ is a tuning hyper-parameter. Setting the gradient to zero yields the following expression for the regression coefficients:

$$w = (X^T X + \lambda I_d)^{-1} (X^T y).$$

Is $(X^T X + \lambda I_d)$ always invertible? If yes, prove it. Otherwise, what conditions should X or λ have?

1.10 Multi-Class Classification with Softmax (3 marks)

Consider the dataset below, which has $n = 10$ training examples, $d = 2$ features, and $k = 3$ classes:

$$X = \begin{bmatrix} 0 & 1 \\ 1 & 0 \\ 1 & 0 \\ 1 & 1 \\ 1 & 1 \\ 0 & 0 \\ 1 & 0 \\ 1 & 0 \\ 1 & 1 \\ 1 & 0 \end{bmatrix}, y = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 2 \\ 2 \\ 2 \\ 2 \\ 3 \\ 3 \\ 3 \end{bmatrix}$$

Suppose we fit a multi-class linear classifier using the softmax loss, and we obtain the following weight matrix:

$$W = \begin{bmatrix} +2 & -1 \\ +2 & -2 \\ +3 & -1 \end{bmatrix}$$

Under this model, what class label would we assign to $\hat{x} = [1, 1]$? (Show your work.)

1.11 Multi-Class Logistic Regression (6 marks)

Assuming that we have n training examples, d features, k classes, t testing examples, and T iterations of gradient descent for training:

- In $O()$ notation, what is the cost of training the softmax classifier with gradient descent?
- What is the cost of classifying the t test examples?

1.1 $A \in \mathbb{R}^{m \times n}$, $B \in \mathbb{R}^{n \times m}$

A and A^T share the same eigenvalues if $m=n$

if $m=n$, A and A^T are both square matrices

let λ_1 be matrix of eigenvalues for A and λ_2 be matrix of eigenvalues for B

$$\det(A - \lambda_1 I) = 0 \quad \det(A^T - \lambda_2 I) = 0$$

using the property $\det(M) = \det(M^T)$ for square matrix M

$$\left(\begin{array}{l} \text{if } M = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \rightarrow M - \lambda I = \begin{bmatrix} a-\lambda & b \\ c & d-\lambda \end{bmatrix} \rightarrow \det(M - \lambda I) = (a-\lambda)(d-\lambda) - bc \\ M^T = \begin{bmatrix} a & c \\ b & d \end{bmatrix} \rightarrow (M - \lambda I)^T = M^T - \lambda I = \begin{bmatrix} a-\lambda & c \\ b & d-\lambda \end{bmatrix} \rightarrow \det(M^T - \lambda I) = (a-\lambda)(d-\lambda) - bc = \det(M - \lambda I) \end{array} \right)$$

$$(A - \lambda I) = (A - \lambda I)^T = A^T - \lambda I^T = (A^T - \lambda I)$$

$\therefore$ Characteristic polynomials are the same, and eigenvalues are the same

$$AB \in \mathbb{R}^{m \times m} \quad BA \in \mathbb{R}^{n \times n}$$

$$\det(AB - \lambda_1 I) = \det(BA - \lambda_2 I) = 0$$

AB and BA share the same set of eigenvalues

$\lambda = 0$ case

$$\text{if } \lambda_1 = 0 \text{ eigenvalue of } AB \quad \det(AB - \lambda_1 I) = \det(AB) = 0$$

$$\det(AB) = \det(A) \cdot \det(B) = \det(BA) = \det(BA - \lambda_2 I) = 0 \quad \therefore \lambda_2 = \lambda_1 = 0$$

$\lambda \neq 0$ case

$$(AB)x = \lambda x \rightarrow \text{since } \lambda \text{ is non-zero, } x \neq 0 \quad (\lambda \text{ is a non-zero eigenvalue of } AB)$$

$$B(AB)x = B\lambda x$$

$$(BA)(Bx) = \lambda(Bx) \rightarrow Bx \neq 0 \text{ because } (AB)x = A(Bx) = \lambda x \neq 0 \text{ (since } \lambda \neq 0 \text{ and } x \neq 0)$$

$\therefore \lambda$ is a non-zero eigenvalue of BA

1.2 $A \in \mathbb{R}^n$ full rank $\rightarrow A^{-1}$ exists...

$$\text{suppose that } Bx = \lambda x \rightarrow x \neq 0 \quad (B \text{ has an eigenvalue } \lambda \text{ and eigenvector } x)$$

$$y = A^{-1}x \rightarrow x = Ay \quad (\text{since } x \neq 0 \text{ and } A^{-1} \text{ exists, } y \neq 0)$$

$$Bx = B(Ay) = \lambda x = \lambda(Ay)$$

$$A^{-1}BAy = A^{-1}\lambda(Ay) = \lambda(A^{-1}A)y = \lambda y \quad (y \neq 0 \text{ and an eigenvector of } A^{-1}BA \text{ with eigenvalue } \lambda)$$

$\therefore A^{-1}BA$ and B both share eigenvalue λ

1.3 $x^T A x > 0$ for $x \neq 0$ and $x \in \mathbb{R}^n$ (A is symmetric $A = A^T$, real eigenvalues)

x is a non-zero eigen vector of A with eigenvalue λ

$$Ax = \lambda x \quad \text{where } x \neq 0$$

$$(x^T)Ax = (x^T)\lambda x = \lambda(x^T x) = \lambda \|x\|^2$$

we know that $\|x\|^2 > 0$ and A is PD $\rightarrow x^T A x > 0$ for $x \neq 0$

$$x^T A x > 0 \rightarrow \lambda \underbrace{x^T x}_{\|x\|^2 > 0} > 0 \quad \therefore \lambda > 0$$

Since $\lambda \neq 0$ (square matrix not invertible if $\lambda = 0$)

since A^{-1} exists, A is full rank

1.4 G is always PSD

$$A \in \mathbb{R}^{m \times n} \quad G = A^T A \in \mathbb{R}^{n \times n} \quad G^T = (A^T A)^T = (A^T)^T A = A \quad (\text{symmetric matrix})$$

$$x \in \mathbb{R}^n: \quad x^T G x = x^T (A^T A) x = (x^T A^T)(Ax) = (Ax)^T (Ax) = \|Ax\|^2 \geq 0 \quad \text{squared norm } \geq 0$$

$$\therefore x^T G x \geq 0, \quad G \text{ is PSD}$$

G is PD if $m \geq n$ and A is full rank

A is full rank $\rightarrow \text{rank}(A) = n$ (linearly independent columns)

$$Ax = 0 \rightarrow x = 0$$

for $x \neq 0$ nullspace of A contains only 0

$$Ax \neq 0 \rightarrow \|Ax\|^2 > 0$$

$$x^T G x = \|Ax\|^2 > 0$$

$$\therefore x^T G x > 0 \rightarrow G \text{ is PD}$$

$$1.5 \quad f(x,y) = \begin{bmatrix} f_1(x,y) \\ f_2(x,y) \\ f_3(x,y) \end{bmatrix} = \begin{bmatrix} x^2 y^3 \\ 4x^2 + \cos(y) \\ 4y^2 - 2x^2 \end{bmatrix}$$

Dimensions of Jacobian of f

$$J_f = \begin{bmatrix} \frac{\partial}{\partial x} f_1 & \frac{\partial}{\partial y} f_1 \\ \frac{\partial}{\partial x} f_2 & \frac{\partial}{\partial y} f_2 \\ \frac{\partial}{\partial x} f_3 & \frac{\partial}{\partial y} f_3 \end{bmatrix} \quad \begin{array}{l} \text{we have 2 inputs (x,y) and 3 outputs (f}_1, f_2, f_3) \\ \therefore \text{Jacobian matrix is 3 rows by 2 columns} \end{array}$$

what is $\frac{\partial}{\partial y} f_2$

$$\frac{\partial}{\partial y} f_2 = \frac{\partial}{\partial y} [4x^2 + \cos(y)] = -\sin(y)$$

what is $\frac{\partial^2}{\partial x \partial y} f_1$

$$\frac{\partial^2}{\partial x \partial y} f_1 = \frac{\partial}{\partial x} \left[\frac{\partial}{\partial y} f_1 \right] = \frac{\partial}{\partial x} \left\{ \frac{\partial}{\partial y} [x^2 y^3] \right\} = \frac{\partial}{\partial x} [3x^2 y^2] = 6xy^2$$

$$1.6 \quad A \in \mathbb{R}^{m \times n}, B \in \mathbb{R}^{n \times m}, (AB + \lambda I_m)^{-1}$$

"square matrix invertible if nullspace contains only $\vec{0}$ "

$$(BA + \lambda I_n)x = 0 \dots \textcircled{D}$$

$$BAx + \lambda I_n x = BAx + \lambda x = 0$$

$$y = Ax$$

$$A(BAx + \lambda x) = (AB)(Ax) + \lambda(Ax) = 0$$

$$AB y + \lambda y = (AB + \lambda I_m)y = 0 \rightarrow y = 0 \quad \text{invertible, nullspace contains only } \vec{0}$$

$$y = Ax = 0 \rightarrow Ax = 0$$

$$\textcircled{D}: (BA + \lambda I_n)x = BAx + \lambda x = \underbrace{B(Ax)}_{=0} + \lambda x = 0 \rightarrow \lambda x = 0$$

$$\lambda \neq 0 \therefore x = 0$$

$\therefore$ nullspace of $\lambda I_n + BA$ contains $\vec{0}$, $(\lambda I_n + BA)$ is invertible

1.7 The correct answer is C, you cannot determine the importance without additional information...

A negative feature constant means predicted output decreases with an increase in that feature.

However, if that feature isn't present then the predicted output wouldn't be significant. It depends on the data in.

$$1.8 \quad \hat{y} = x_j w_j \quad x_j \in \mathbb{R}^n \text{ and } w_j \in \mathbb{R}^1$$

$$r = \|y - \hat{y}\|^2 = \|y - x_j w_j\|^2$$

$$r = \|z\|^2 = \vec{z}^T \vec{z} = (y - x_j w_j)^T (y - x_j w_j) = (y^T - w_j x_j^T)(y - x_j w_j) = y^T y - y^T x_j w_j - w_j x_j^T y + w_j^2 x_j^T x_j = \|y\|^2 - 2w_j y^T x_j + w_j^2 \|x_j\|^2$$

$$\frac{dr}{dw_j} = -2y^T x_j + 2w_j \|x_j\|^2 = -2(y^T x_j - w_j \|x_j\|^2)$$

$$\frac{dr}{dw_j} = 0 \rightarrow y^T x_j - w_j \|x_j\|^2 = 0 \rightarrow w_j = \frac{y^T x_j}{\|x_j\|^2} = \frac{x_j^T y}{x_j^T x_j}$$

$$\therefore w_j = \frac{x_j^T y}{x_j^T x_j}$$

1.9 is $(x^T x + \lambda I_d)$ always invertible? with the right λ , yes... (this is the whole point of ridge regression...)

$$z = (x^T x + \lambda I_d) x, x \in \mathbb{R}^{n \times d}, z \in \mathbb{R}^{d \times d}$$

"square matrix invertible if nullspace contains only $\vec{0}$ "

$x^T x$ is always PSD ($a^T x^T x a = \|x a\|^2 \geq 0$)

$$a^T z a = a^T (x^T x + \lambda I_d) a = a^T x^T x a + \lambda a^T a = \|x a\|^2 + \lambda \|a\|^2$$

if $a \neq 0$, $\lambda > 0$ to make this PD (invertible)

$\rightarrow x^T x + \lambda I_d$ is PD if $\lambda > 0$

if $a \neq 0$ & $\lambda = 0$, $a^T x^T x a = \|x a\|^2 > 0$ which means $x^T x$ has to be PD

to make x PD, $\text{rank}(x) = d$ such that $x b \neq 0$ for $b \neq 0$

$\therefore$ if $\lambda > 0$, x can be less than full rank but

if $\lambda = 0$, x has to be full rank ($\text{rank}(x) = d$)

$$1.10 \quad S = W \hat{X} = \begin{bmatrix} 2 & -1 \\ 2 & -2 \\ 3 & -1 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 2 \end{bmatrix}$$

$$\text{Softmax probabilities } P(y=j|x) = \frac{e^{s_j}}{\sum_{k=1}^3 e^{s_k}}$$

$$P(1) = \frac{e^1}{e^1 + e^0 + e^2}, P(2) = \frac{e^0}{e^1 + e^0 + e^2}, P(3) = \frac{e^2}{e^1 + e^0 + e^2}$$

$\therefore$ Since $e^2 > e^1 > e^0$ $j=3$ (S_3) is the largest probability and would get

be the class label for $\hat{X} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$

1.11

n training examples t testing samples

d features T iterations

K classes

Weight matrix $W \in \mathbb{R}^{d \times K}$ and for each training iteration T we need to compute the class scores for n examples $S = W^T X + b$ ($W \in \mathbb{R}^{d \times K}$ $X \in \mathbb{R}^{d \times n}$ $S \in \mathbb{R}^{K \times n}$) adding in bias $b \in \mathbb{R}^{K \times n}$ $\rightarrow$ cost so far: $O(nkd) + O(nK)$ applying softmax: $O(nK)$ gradient $\nabla_{w_i} = \frac{1}{n} \sum_{i=1}^n (\hat{y}_i - y_i) x_i$ $\frac{\partial L}{\partial b} = \frac{1}{n} \sum_{i=1}^n (\hat{y}_i - y_i)$ gradient descent iteration: $O(nkd)$ (according to chat GPT) $\therefore$ for 1 iteration $\rightarrow O(nkd) + O(nK) + O(nK) + O(nkd) \simeq O(nkd)$ (dominant order, as d grows, nK drops out)and for T iterations $\rightarrow O(Tnkd)$

testing and classification is just 1 iteration and doesn't involve gradient descent

 $O(nkd) + O(nK) \simeq O(nkd)$ $\therefore$ Cost $\Rightarrow O(nkd)$